OneOhm — Final Product Requirements & Functional Goals (v2)

Oct 31, 2026 · @Tejas Patil

What changed in v2

This version is the single baseline for building OneOhm in one go. It keeps every requirement from the previous version and resolves the gaps found in review.

Main objective (unchanged): build a simple, premium and intelligent business operating system that manages OneOhm's entire business with minimum manual effort, maximum visibility and no unnecessary complexity.

Area	Decision in v2	Section
Government subsidy	Shown in every quotation, calculated automatically from a configurable subsidy rule table. OneOhm does not pay the subsidy.	3.5, 3.11
DCR / non-DCR panels	Every panel is tagged. Subsidy applies only with DCR panels. Optional side-by-side comparison in the quotation.	3.5
Loans	Payer source per milestone, separate bank receivables, disbursement tranches linked to tasks.	3.11, Flow B
Subcontractors	Contractor profiles, crews, rate cards, work orders, payables and performance, using the same assignment engine.	3.13, 4.9
Quick quote vs locked quote	One quotation flow with smart defaults, built-in contingency and a survey tolerance band.	3.5, Flow F
DISCOM	DISCOM master with per-DISCOM task templates. Selected at project creation. DISCOM delays tracked separately.	3.16

Area	Decision in v2	Section
Final payment trigger	Final payment becomes due at installation and commissioning. Net-meter completion closes the project operationally. Trigger is configurable.	3.7, Flow D
Motion and 3D	No generated videos. Focus is UI feel and experience.	9
Field uploads	Online-only, with automatic retry of evidence uploads.	4.4, 10
Delivery	Entire system is built together, not in phases.	12

Where a detail is not written here, it is resolved during implementation so the system stays self-handling, connected and easy to use.

1. Non-negotiable product goals

Priority	Goal	Required outcome
1	Exceptional UI/UX	Beautiful, calm, premium and extremely easy to use
2	Zero-learning-curve experience	Employees, contractors and owners can use common workflows without training
3	Complete business control	Every activity, payment, task and responsibility is traceable, including contractor work
4	Smart automation	System handles task generation, assignment, reminders and follow-ups
5	Operational accuracy	Pricing, subsidy, material, labor, payments and project status stay connected
6	Customer experience	Customers always know their project status, payments, subsidy and next steps
7	Configurability	Owners can change normal business rules without developers
8	Simplicity	No unnecessary features, screens, settings or complicated workflows

Explicit product decisions

- Web application for owners, managers and office operations.
- Employee, contractor and reseller mobile application (one app, role-based home screens).
- Separate customer mobile application.
- OTP login for mobile apps; ID/email and password for web.
- Online-only operation. No offline mode or offline synchronization. Evidence uploads retry automatically until they succeed.
- No unnecessary document version-history system.
- Documents organized by project, category and source.
- Task-level customer WhatsApp notifications are configurable.
- Government subsidy is shown in quotations but is never paid or deducted by OneOhm.

- All project, task, finance, inventory, contractor and customer workflows share one source of truth.
- Motion and interaction quality is a core design requirement, but never compromises usability or performance. No generated videos.
- The entire system is built and released together.

2. Product platforms

Platform	Used by	Covers
Business web app	Owners, managers, office team	Owner dashboard, CRM, quotations, projects, finance, operations, employees, contractors, DISCOM, configuration and reports
Field mobile app	Employees, contractor crews, resellers	Leads, surveys, quotations, site activities, tasks, attendance, leave, rewards, contractor work orders and reseller earnings
Customer mobile app	Customers	Project journey, payments, subsidy information, documents, service, feedback, referrals and rewards
Shared business platform	All apps	Common records, pricing and subsidy engine, task engine, workflow, notifications, permissions, finance and analytics

3. Business web app — module requirements

3.1 Owner dashboard and business command center

Goal: let the owner understand the complete business in one glance and focus only on exceptions that need attention.

- Active, completed, delayed, blocked and at-risk projects.
- Leads, sales pipeline, quotations and conversion.

- Revenue, collections, receivables (customer and bank) and payables (vendor and contractor).
- Employee and contractor availability, attendance, leave and capacity.
- Project health and delivery forecasts, using defined, deterministic formulas.
- Delays split by cause: internal, customer, bank or DISCOM.
- Pending approvals and decisions.
- Customer complaints and service requests.
- Material shortages and procurement risks.
- Overdue follow-ups and unresolved blockers.
- Project-wise, department-wise, employee-wise and contractor-wise drill-down.
- Daily and weekly business summaries.

Requirement: every metric opens the records behind it. No decorative numbers without an actionable explanation.

3.2 Customer and property management

Goal: maintain one customer identity and a complete history of their relationship with OneOhm.

- Individual and organization profiles, including housing societies.
- Multiple properties and projects per customer.
- Customer, consumer and electricity details, including the connected DISCOM and meter holder name.
- Customer category for subsidy: individual residential, housing society (with number of households), or commercial.
- Address, GPS, site photographs and documents.
- Communication and follow-up history.
- Quotations, projects, payments and outstanding amounts.
- Change requests and approvals.
- Service, warranty and AMC history.
- Customer feedback and referral activity.
- Complete Customer 360 view.

Avoid duplicate identities when one person is also a reseller, employee, contractor or other business contact.

3.3 Lead and sales management

Goal: every lead has an owner, a next action and a traceable conversion journey.

- Manual, website, WhatsApp, employee and reseller lead capture.
- Configurable lead stages.
- Lead assignment and follow-up scheduling.
- Call, visit and communication history.
- Customer and property matching.
- Duplicate detection.
- Qualification and conversion.
- Lost reasons and future follow-up.
- Salesperson and reseller performance.

Rule: an open lead cannot silently remain without a next action.

3.4 Site survey and feasibility

Goal: capture accurate site information once and reuse it in design, quotation and project execution.

- Survey scheduling and assignment.
- Residential, housing society, commercial, rooftop and ground-mount templates.
- Site dimensions and usable area.
- GPS check-in and geotagged photographs.
- Electricity bill, consumer number, DISCOM and meter holder name.
- Roof, structure, shadow and installation details.
- Floor-wise and route-wise cable measurements.
- Survey checklist and required evidence.
- Voice notes and measurements.
- Survey report generation and approval.
- On submission, the quotation cost is recalculated automatically against the survey (see 3.5 tolerance band).

3.5 Product catalog, quotation and subsidy

Goal: generate an accurate, attractive quotation in a few taps, showing the customer exactly what they pay and how much government subsidy they can expect.

Catalog and quotation features

- Solar panels, inverters, batteries, structures, electrical equipment, cables and accessories.
- Product specifications, brand, SKU and unit.
- Every panel tagged DCR or non-DCR.
- Purchase price, selling price and tax configuration.
- Installation, transport and service charges.
- Discounts, gifts and optional products.
- Survey-linked capacity and system design.
- Automatic BOM and cost calculations from the central cost engine (3.9).
- Automatic subsidy calculation.
- Quotation revisions and approvals.
- PDF generation, WhatsApp sharing, download and email.
- Customer acceptance and rejection tracking.
- Accepted quotation converted into project.

Subsidy calculation

OneOhm does not pay the subsidy. It is credited by the government to the customer's bank account after commissioning. OneOhm invoices and collects the full project price; the quotation only shows the expected subsidy.

Initial subsidy rules under PM Surya Ghar (as of October 2026). All values live in a configurable subsidy rule table with valid-from dates, so changes need no developer.

Customer category	Rule	Maximum
Individual residential	₹30,000 per kW for the first 2 kW, ₹18,000 for the 3rd kW	₹78,000 (3 kW and above)
Housing society / RWA, common facilities	₹18,000 per kW, eligible capacity = lower of (households × 3 kW) or installed capacity, up to 500 kW	₹90,00,000
Commercial / industrial	Not eligible	₹0
Maharashtra state top-up	No general top-up; separate scheme for low-income households only	Configurable

Eligibility checks applied automatically:

- Panels must be DCR. Non-DCR panels set the subsidy to ₹0 with a one-line reason.
- On-grid or hybrid systems only.
- Housing society meter must be in the society's name, not the builder's.
- One subsidy per property.

Proposal flow (no extra steps)

1. Sales selects or confirms the customer category, usually pre-filled from the property.
2. Sales selects system size and panel product.
3. The system calculates price, subsidy and effective cost instantly.
4. Optional: show a DCR vs non-DCR side-by-side (price, subsidy, effective cost) so the customer decides in seconds.
5. The PDF shows: Total project price → Expected government subsidy → Effective cost to customer, with a short note that the subsidy is credited by the government after commissioning.

The subsidy calculation is saved in the accepted quotation snapshot. Later rule changes never alter an accepted quotation; the system flags affected open quotations instead.

Quick quote with tolerance band

One quotation flow, fast and still accurate:

1. The cost engine fills cable lengths, structure and labor from configured defaults per site type (floor count, roof type, structure type). No measurement is needed to quote.
2. Each default-based component carries a configurable contingency.
3. The quotation can be accepted immediately.
4. The site survey captures actual measurements and the system recalculates automatically.
5. Cost difference within the configured tolerance: absorbed. The customer sees no change; management sees the variance.
6. Cost difference beyond tolerance: a revision is generated for manager approval and customer re-confirmation.
7. The project budget locks only after survey verification.

Rule: an accepted quotation revision is locked. Any later commercial or scope change requires a new revision and approval.

Sources: PM Surya Ghar CFA structure (JBVNL) · PM Surya Ghar consumer manual (CAG) · MNRE amended guidelines on DCR · Maharashtra state subsidy status

3.6 Project management

Goal: manage the complete project journey from accepted quotation through commissioning, handover and financial closure.

- Residential, housing society and commercial project types.
- Project creation from accepted quotation.
- DISCOM selection at project creation (3.16).
- Project templates and conditional tasks.
- Project manager, team and contractor selection.
- Automatic task generation and role-based assignment.
- Milestones, deadlines and dependencies.
- Project cost and accepted scope.
- Project health and risks, with delay cause tracking.
- Customer-visible milestones.
- Project change requests.
- Documents, activity history and approvals.
- Installation, commissioning, DISCOM and net-metering stages.
- Subsidy status tracking for customer visibility.

- Project completion and financial closure tracking.

3.7 Dynamic task template and notification engine

Goal: let the business define exactly what happens at each project task, including responsibilities, customer communication and downstream business actions.

Configuration	Requirement
Task name and description	Simple, understandable task definition
Project conditions	Add only when relevant to project type, DISCOM or selected options
Assignee rules	Assign by role, team, skills and availability; employee or contractor
Dependencies	Define which task must happen first
Required evidence	Photos, documents, measurements or checklist
Approval	Optional manager or designated reviewer
Due date	Relative to project start or another milestone
Waiting party	Internal, customer, bank or DISCOM, for delay attribution
Employee notification	Notify assignee and relevant manager
Customer WhatsApp notification	Enable/disable per task
Customer message	Configurable message template
Customer visibility	Whether task status is visible to customer
Completion action	Trigger next task, milestone, payment milestone, contractor payable or business event
Escalation	Reminders and escalation if overdue or blocked

OneOhm's own task templates, DISCOM templates, rate cards and payment milestones ship pre-configured, so the system works on day one.

Final payment trigger

- Default: the final payment milestone becomes due when installation and commissioning are complete. These are steps OneOhm controls.
- Net-meter installation is a DISCOM step. Its completion sends the customer notification, updates the journey and closes the project operationally. It does not hold the main payment.
- The trigger of every payment milestone is configurable, so the owner can change this rule without developers.

Example — net meter installation task: assigned to the net-metering coordinator, evidence is the installation confirmation or photo, customer WhatsApp enabled, customer-visible. When completed:

1. Completion and evidence are recorded.
2. The configured WhatsApp message is sent and delivery status recorded.
3. Customer-visible progress updates.
4. The project's operational completion conditions are evaluated.
5. If all are met, the project is marked operationally complete.

Project completion and financial settlement remain separate events.

3.8 Automatic project task generation and team assignment

Goal: project creation is fast, accurate and mostly automatic.

The owner only confirms: customer and property, accepted quotation, project type, DISCOM, loan (yes/no and bank), selected options (battery, ground mount, meter change), subsidy applicability, project manager, team and contractors, and target dates where required. The system handles the rest.

Project condition	Automatically generated work
Residential solar	Residential base template
Housing society	Society template, including society documents and meter checks
Commercial solar	Commercial template
Selected DISCOM	That DISCOM's application, inspection and net-meter tasks

Project condition	Automatically generated work
Subsidy applicable	Subsidy portal and document tasks, customer-visible
Loan required	Loan documentation, bank submission, approval and disbursement tasks
Battery included	Battery procurement, installation and testing
Ground mount	Ground structure and foundation
Consumer name change	Document and utility change-request tasks

Owners configure which conditions generate which tasks.

Smart assignment

- Employees and contractor crews have defined roles, skills and capabilities.
- Task templates define the required role or skill.
- The system finds eligible people within the selected team and contractors.
- It considers workload, leave, availability, cost and assignment rules.
- Routine tasks auto-assign under approved rules; critical or unusual ones need manager confirmation.
- Unassigned tasks are clearly highlighted.

Success condition: project creation never requires manually assigning 30–45 tasks one by one.

3.9 Solar costing, estimation and pricing engine

Goal: one structured, configurable and measurable estimation system that is fast for sales and accurate for execution. Per-kW rates remain only as benchmarks and default fallbacks.

A. Electrical cable and wiring

Inputs: inverter location, panel array location, floor number and height, horizontal and vertical route distances, DC/AC cable types and sizes, number of runs, routing method and conduit/tray, earthing, and wastage allowance.

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\text{Cable cost} = \sum (\text{Measured length} \times \text{Rate per meter})
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Different cable types, sizes and routing methods have separate rates. Before the survey, lengths come from site-type defaults.

B. Distance-based travel and labor

- Site distance from the configured base location and per-km travel rates.
- Number of visits, workers, labor category and skill.
- Estimated hours or days, floor-wise extra effort and site accessibility.
- Transport and equipment charges.
- When a contractor is assigned, their rate card is used.
- The same travel distance is never charged twice.

C. Mounting structure

Structure-specific configurations instead of per-kW-only costs: aluminum rail, standard rooftop, ground-mount and custom structures; rail length and quantity; mounting feet, clamps and fasteners; supports and foundations; material specs; height and tilt; fabrication and installation labor; transport. For example, a 3 × 6 aluminum rail arrangement is represented by its actual component quantities and rates.

D. Central cost engine

Flow: survey or site-type defaults → equipment BOM → cable estimate → structure BOM → labor and travel → other charges → project cost (material + labor + travel + structure + overhead + margin) → quotation, subsidy display, BOM, budget and procurement.

- Configurable units and rate cards.
- Contingency per default-based component and survey tolerance band (3.5).
- Manual adjustment with reason and approval rules.
- Estimated vs actual material, labor and contractor cost.
- Cost variance reports and site-specific charges.
- No duplicate pricing calculations across modules.
- Accepted quotation cost and subsidy snapshot retained.
- Rate changes never silently change accepted quotations.

Success condition: sales generates a quotation in minutes, while management sees exactly how the price was calculated.

3.10 Inventory, procurement and vendor management

Goal: connect project BOM, purchasing, stock and vendor expenses.

- Central product and material catalog, with DCR/non-DCR tags on panels.
- Stock by warehouse/location.
- Project-wise reserved material.
- Purchase requests and purchase orders.
- Goods receipt and inspection, including panel serial numbers and DCR certificates where needed.
- Material issue, usage and return, including material issued to contractors.
- Vendor invoices and matching.
- Vendor payments and advances.
- Stock shortage alerts.
- Project material reconciliation.
- Vendor performance and outstanding.

3.11 Finance and payment management

Goal: every rupee received, collected or paid is recorded, allocated and reconciled.

Core finance

- Customer payment milestones with configurable triggers.
- Razorpay payment link creation.
- Partial and multiple payments.
- Cash, cheque and bank transfer records.
- Receipts and invoices for the full project price.
- Project-wise payment allocation.
- Refunds and excess amounts.
- Outstanding and overdue reminders.
- Vendor and contractor payments through configured methods and RazorpayX where eligible.
- Payment approval workflows.

- Gateway fee and settlement reconciliation.
- Cash collection and deposit reconciliation.
- Double-entry financial ledger.
- Tax and TDS configuration, validated with finance professionals.

Loan-funded projects

- Each payment milestone has a payer source: Customer or Bank.
- A loan project has two receivables: the customer's own share and the bank disbursement (single or in tranches).
- Bank disbursements are recorded and reconciled as bank receipts. No Razorpay link is sent for the bank's share.
- Disbursement conditions (documents, completion proof) are linked to project tasks. When they complete, finance is prompted to claim the disbursement.
- Lending partners are configurable, each with required documents and disbursement pattern.
- Clear statuses: "Waiting for bank disbursement" vs "Customer payment pending", on the dashboard and in the customer app.

Subsidy in finance

- The subsidy is never recorded as OneOhm revenue, a discount or a receivable.
- Invoices show the full price; the expected subsidy appears only as information.
- Subsidy status (applied, inspected, released to customer) is tracked for customer visibility only.

Important: project completion may create a final receivable. Payment receipt, project completion and financial closure remain distinct events.

3.12 Employee, leave, capacity and rewards

Goal: keep work owned, employees motivated and projects protected from unexpected absence.

- Employee profiles, roles and skills.
- Attendance and configurable tracking policies.
- Leave balance and mobile leave applications.
- Leave impact assessment, backup recommendations and handover.
- Workload and capacity management.

- Automatic task assignment recommendations.
- Task completion and quality scoring.
- Points, recognition and project bonuses.
- Customer feedback linked to employee performance.
- Approved incentive and payout tracking.
- Delays caused by customers, banks or DISCOMs never count against employee performance.

3.13 Subcontractor and labor crew management

Goal: manage contractor work as cleanly as employee work, so every responsibility stays traceable and paid correctly.

- Contractor profile: company or individual, contact, crew members, skills, service areas, documents and payment terms.
- Rate card: per kW, per job, per day or per task type.
- Same assignment engine: tasks go to employees or eligible contractors by role and skill rules.
- Crew access: crew members log in to the field app with a restricted role and see only their assigned tasks.
- Work order: generated automatically from the tasks assigned to a contractor on a project, shareable as PDF or WhatsApp.
- Evidence and checklists: same requirements as employees.
- Material: material issued to a contractor is tracked and reconciled per project.
- Payables: approved task completion creates a contractor payable in the ledger. Supports advances, rework deductions, retention and TDS.
- Costing link: contractor rates feed estimates; actual payouts feed estimated vs actual and project profitability.
- Performance: quality, delays, rework and customer feedback per contractor.
- Contractors never see pricing, margins, other contractors or internal discussions.

3.14 Documents and approvals

Goal: store all project documents in one easy-to-find location, regardless of where they came from. No complex version-control system.

Sources: customer onboarding, lead capture, employee, contractor or reseller uploads, site surveys, project tasks, system reports, quotations, invoices and receipts, vendor submissions, bank and DISCOM documents, subsidy documents, commissioning and handover.

- Auto-linked to the correct customer, property and project.
- Organized by category and source.
- Search and filter by name, type, date and source.
- Preview common file types; upload and download.
- Required and missing documents marked, per project type, DISCOM, loan and subsidy.
- Print individual documents.
- Select multiple documents and generate a combined PDF or ZIP.
- Simple printable document checklist.
- Control who can view, upload, print and download.
- When a document is replaced, the previous file and who replaced it are kept in the audit trail, without exposing versioning to everyday users.

3.15 Change requests

Goal: track customer, utility, commercial and project changes without losing control of dependencies.

Utility change request types (MSEDCL, current)

Change request	What it does	User input
Consumer Name Change	Transfers the electricity connection to a new owner's name	New consumer name (typed)
Name Spelling Correction	Corrects a spelling error in the consumer name on the MSEDCL record	Correct spelling (typed)
Property Type Change	Changes the connection category, e.g. residential to commercial	New property type (selected)
Load Change	Increases or decreases the sanctioned load	Phase: single or three (selected); new load in kW (typed)

Change request	What it does	User input
New EV Meter	Requests a new meter for electric vehicle charging	Optional note
New Connection	Requests a new electricity connection for the site	Phase (selected); optional note

Types, their input fields, required documents and task templates are configurable per DISCOM, so new types or other DISCOMs' processes can be added without system changes.

How change requests are managed

1. **Created in one step** from the customer, property or project, by picking the type and filling only its fields above.
2. **Suggested automatically** when the system detects a mismatch, for example:
   - Meter holder name differs from the customer name → suggest Consumer Name Change or Name Spelling Correction.
   - Proposed solar capacity exceeds the sanctioned load → suggest Load Change.
   - Site has no connection → suggest New Connection.
3. **Task set generated** from the type's DISCOM template and assigned to the coordinator by the normal assignment rules.
4. **Tracked as waiting on DISCOM**, so utility time is reported separately from internal delays.
5. **Master data updated automatically on completion**: consumer name, property type, phase, sanctioned load, and new meter or connection details on the property. The previous value is kept in history.
6. **Customer kept informed** through configurable WhatsApp notifications and a simple milestone in the customer app.

Project impact (default rules, configurable)

Change request	Effect on project
Consumer Name Change / Name Spelling Correction	Blocks net-metering and subsidy tasks until completed, since documents must match the consumer name

Change request	Effect on project
Property Type Change	Re-checks subsidy eligibility; a change to commercial removes the subsidy and flags open quotations for revision
Load Change	Blocks net-metering application until the sanctioned load supports the installed capacity
New Connection	Blocks commissioning and net-metering tasks until the connection is active
New EV Meter	No blocking; tracked as an independent request

Only genuinely dependent tasks are blocked; all other project work continues.

Other change requests

- Project scope and product changes, including panel type changes that affect subsidy.
- Timeline, cost and subsidy impact shown before approval.
- Approval and customer confirmation.

Change requests can exist with or without a solar project, so existing customers can raise them on their own.

3.16 DISCOM management

Goal: handle every DISCOM's process cleanly, while keeping utility delays visible and separate from OneOhm's performance.

- DISCOM master: register DISCOMs (for example MSEDCL, Tata Power, Adani, BEST) with their task template, required documents and expected timelines.
- Project creation: one dropdown to select the DISCOM, pre-suggested from the property's consumer details.
- The selected DISCOM's task set is added automatically.
- DISCOM tasks are completed by the coordinator once the DISCOM finishes its side, with evidence.
- Waiting-on-DISCOM time is shown separately on the project and dashboard.

- Average time per DISCOM step is tracked, so the business can see which DISCOM or office causes delays.
- Customer sees DISCOM steps as simple milestones (application, inspection, net meter installed).

3.17 Commissioning, handover and after-sales

Goal: every completed project has the required evidence, customer handover and future service records.

- Installation checklist.
- Testing and commissioning, which triggers the final payment milestone by default.
- DISCOM inspection and net-meter status.
- Equipment serial numbers.
- Final material reconciliation, including contractor-issued material.
- Required reports and certificates, including subsidy-related documents.
- Customer sign-off and handover package.
- Warranty and installed asset records.
- Service requests, AMC and preventive maintenance.

3.18 System-generated project reports

Goal: every statutory and DISCOM report is generated as a PDF from project data, with no re-typing and no incomplete documents.

The existing report templates and `definitions.ts` will be provided at implementation time and are the source of truth for layouts and fields. Triggers below are defaults and configurable.

Report	Purpose	Default trigger	Condition	Key data
Work Completion Report (WCR)	Certifies the solar plant installation is complete (2 pages)	Installation completed	Always	Panel, inverter, earthing and capacity details

Report	Purpose	Default trigger	Condition	Key data
Annexure-I & Proforma-A	MSEDCL commissioning report for a grid-connected plant	Commissioning completed	MSEDCL projects	Consumer, sanctioned kW, installed kW, mount type, inverter and panel details
Net Metering Agreement	Agreement between consumer and MSEDCL for net metering; no signatures, as submission is the agreement	Net-metering application stage	MSEDCL projects with net metering	Consumer and system details
DCR Undertaking / Self-Declaration	Company declares panels are made in India	Subsidy document stage	Required only when subsidy applies; otherwise shows "Not needed"	Panel make, model and serials
Model Agreement	MNRE model agreement between customer and vendor	Project onboarding	Always (confirm current format at implementation)	System, cost and project onboarding date
Meter Test Covering Letter	Requests MSEDCL to test the generation meter	Generation meter details captured	MSEDCL projects	Meter serial number, make and capacity

Rules

- Every report field is mapped to the task that captures it, and that task requires the field as evidence. For example, the meter installation task requires meter serial, make and capacity.
- Data is captured once and reused across reports, quotations and documents.

- Before generating, the system checks all required fields. If any are missing, it shows exactly which field and which task captures it, and does not produce an incomplete PDF.
- The DCR undertaking follows the same subsidy flag used by the quotation and tasks.
- Report templates are linked per DISCOM, so other DISCOMs' formats can be added without system changes.
- Generated reports are saved automatically to the project documents (source: System-generated), regenerated if source data changes before submission, and available for single or combined printing.
- Visibility per report (internal or customer-visible) is configurable.

Acceptance: for a completed MSEDCL subsidy project, all six reports generate correctly from project data in one action; for a non-subsidy project, the DCR undertaking shows "Not needed"; a missing field blocks generation with a clear message.

4. Employee, contractor and reseller mobile app

Goal: users open the app, immediately understand their work and complete it with minimal effort. The app must not feel like an ERP.

4.1 Employee home screen

- Today's assigned tasks and upcoming site visits.
- Tasks requiring updates, blocked tasks and help requests.
- Attendance status, leave balance and quick leave application.
- Points and rewards.
- Pending uploads indicator.
- Important announcements and role-specific quick actions.

4.2 Site activity and survey

- New lead onboarding at the site.
- Pre-project survey and post-project inspection.
- GPS check-in and check-out.
- Geotagged photos and videos.
- Measurements and observations.
- Electricity bill, consumer number and DISCOM capture.

- Voice notes, survey checklist and submission.
- Project-linked activity history.

4.3 Attendance and authorized location tracking

- OTP-based login.
- Configurable attendance marking, with optional automatic capture.
- Configurable day-start and day-end policies.
- Location tracking configurable by employee, role and working hours.
- Coordinates and timestamps when enabled; battery and connectivity info where the OS permits.
- Site check-ins, travel records, distance and reimbursement.
- Attendance correction requests.
- Explicit consent captured in the app, a visible tracking status and a configurable data-retention period.

No hidden or 24-hour tracking.

4.4 Task execution

- Start, update, block and complete tasks.
- Task-specific checklists, required photos and documents.
- Voice-to-text progress notes.
- Reason for blocked work and assistance requests.
- Submit for approval; see feedback and corrections.
- Reminders and escalations.
- Evidence is saved on the device and uploaded with automatic retry until it succeeds. A task completes only after its required evidence is uploaded.

4.5 Lead and quotation creation

- Capture new leads; find existing customers; create properties.
- Capture documents and electricity bills.
- Conduct site surveys.
- Create quotations with the central pricing and subsidy engine, including the DCR vs non-DCR comparison.
- Request discount approval.

- Generate PDF; share through WhatsApp or download.
- Track customer response.

4.6 Leave and handover

- View leave balance and apply in a few steps.
- See affected projects, tasks and suggested backups.
- Hand over tasks and pending information.
- Approval, status, history and emergency leave.

4.7 Employee engagement

- Personal points, achievements and customer feedback.
- Team recognition and project bonuses.
- Reward history and approved incentive payments.

4.8 Reseller experience

- Register and manage leads.
- View their customers, projects and progress.
- Generate and share authorized quotations.
- View commission rules; track earned, pending and approved commission.
- Payment history, revenue statements, notifications and support.

Resellers never see internal discussions, private financial data or other resellers' leads.

4.9 Contractor crew experience

- Home screen with today's assigned tasks and sites.
- Work orders per project.
- Task execution with the same checklists and evidence as employees.
- Material received and returned per project.
- Approved work, pending payables and payment history.
- Ratings and rework feedback.

Contractors see only their own tasks, work orders and payments.

5. Customer mobile app

Goal: a premium, transparent and rewarding experience through the whole solar project lifecycle.

5.1 Home and account

- OTP-based login and profile.
- Multiple properties and projects.
- Project overview, pending customer actions and notifications.
- Preferred language.
- Authorized family or contact access where supported.

5.2 Project journey

- Visual project progress with completed and upcoming milestones.
- Survey, design, material and installation updates.
- DISCOM application, inspection and net-meter progress.
- Subsidy status: expected amount, applied, inspected, released by government.
- Commissioning and completion.
- Photos, reports and next expected activity.
- Delay notification with the reason (for example, waiting on DISCOM).
- Project team contact information.

Customers see understandable milestones, never internal task terminology.

5.3 Payments

- Accepted quotation, project price, expected subsidy and effective cost.
- Payment milestones, paid and outstanding amounts.
- For loan projects: own share vs bank disbursement status.
- Razorpay links, UPI and other supported methods.
- Receipts, invoices, payment history and refund tracking.
- Clear payment purpose for every milestone.

5.4 Documents

- Quotations, survey reports, invoices and receipts.

- Installation and commissioning reports.
- Warranty certificates, product manuals and handover documents.
- Subsidy and DISCOM documents relevant to the customer.
- Upload requested documents.

5.5 Service and support

- Create service requests for a project or installed system.
- Upload photos and videos; describe the issue or use voice input.
- Track ticket status and scheduled visits.
- Warranty and AMC details.
- Support, escalation and rating after resolution.

5.6 Refer and earn

- Personal referral link/code and WhatsApp sharing.
- Referral registration and progress.
- Configurable reward amount and conditions.
- Pending, approved and paid rewards, history and notifications.

The owner configures qualifying events, campaign budgets, reward limits and eligibility.

5.7 Feedback and ratings

- Rate the project experience and stages such as survey, installation and support.
- Optionally rate individual team members or contractor crews.
- Comments, submitted after meaningful milestones, with acknowledgement.

Verified feedback feeds employee and contractor performance, with safeguards against unfair or manipulated ratings.

6. WhatsApp chatbot and communication

Goal: OneOhm is accessible even to customers who don’t install an app or open a website.

6.1 Customer WhatsApp assistant

- Verify customer identity.
- Show project progress, upcoming milestones and subsidy status.

- Share payment links and payment status.
- Send quotations, invoices and receipts.
- Request or receive documents.
- Create and track service requests.
- Share warranty and AMC information.
- Capture feedback, register referrals and show reward status.
- Escalate to a human team member.

6.2 Task-triggered WhatsApp notifications

Each task template has its own customer communication settings:

- Enabled/disabled; customer-visible or internal-only.
- Trigger event: started, completed, approved or another milestone.
- Message template, recipient and language.
- Required data fields and optional attachments or report links.
- Delivery tracking, retry and failure handling.

Example events: survey completed, design approved, material delivered, installation started, installation completed, commissioning completed (final payment due), DISCOM inspection done, net meter installed, project completed, payment milestone due.

Rule: internal tasks never automatically become customer notifications. Only explicitly configured tasks and events communicate with customers.

7. Shared automation and configuration

Goal: the system adapts to OneOhm's changing operations without custom development for routine business changes.

7.1 Workflow engine

- Event-based triggers and configurable conditions.
- Task generation and assignment rules (employees and contractors).
- Customer and employee notifications.
- Approval workflows, reminders and escalations.
- Project milestone transitions.
- Payment receivable creation (customer and bank) and contractor payable creation.

- Workflow history and failure visibility.

7.2 Central configuration

Authorized owners/admins configure:

- Lead stages and project types.
- Task templates, conditional generation and notification rules.
- Roles, skills and assignment rules.
- Approval policies.
- Leave, attendance and location-tracking rules.
- Rewards, points and referral programs.
- Product catalog, DCR tags and rates.
- Cable, structure, labor and travel rate cards; site-type defaults, contingency and survey tolerance.
- Subsidy rule table (rates, slabs, caps, eligibility, valid-from dates).
- DISCOM master, DISCOM task templates and DISCOM report templates.
- Lending partners and disbursement patterns.
- Contractors and rate cards.
- Payment milestones and their triggers.
- Document requirements and company profile (legal name, addresses, GSTIN, contacts, logo).
- WhatsApp templates.
- Service categories and SLAs.
- Dashboard settings.

Configuration uses simple forms, clear labels, defaults and previews. Everything ships pre-configured with OneOhm's current processes. Ordinary users never edit code or raw rule expressions.

7.3 Permissions and audit

- Role-based and permission-based access with configurable business roles.
- Customer, reseller and contractor data isolation.
- Web password and mobile OTP authentication.
- Sensitive action approvals.

- Audit history for important changes, including document replacements.
- Login and security activity.
- Secure document access.
- Financial action traceability.

8. End-to-end business flows

The system is designed around complete journeys, not isolated screens.

Flow A — Lead to project

1. Lead received; customer matched or created; property linked.
2. Follow-up scheduled.
3. Quick quotation created from site-type defaults, with subsidy and effective cost shown.
4. Customer accepts quotation.
5. Site survey completed; cost recalculated against the tolerance band.
6. Within tolerance: budget locks. Beyond tolerance: revision approved and re-confirmed by customer.
7. Project created; DISCOM, loan and options confirmed.
8. Conditional tasks generated; team and contractors assigned automatically.
9. Payment schedule activated; execution begins.

Flow B — Loan-funded project

1. Loan required = yes; lending partner selected.
2. Loan task set added; finance roles assigned.
3. Customer share and bank share set as separate receivables.
4. Documents checked; bank submission and approval tracked.
5. Disbursement conditions met by linked tasks; finance prompted to claim.
6. Each disbursement recorded and reconciled.
7. Only tasks that genuinely depend on the loan wait for it.

Flow C — Task completion to customer communication

1. Employee or contractor completes task; evidence uploaded.

2. Manager approval if configured.
3. Task marked complete.
4. Customer WhatsApp notification if enabled.
5. Customer journey updated; next dependent task unlocked.
6. Contractor payable created if applicable.
7. Project health recalculated.

Flow D — Commissioning to final collection and closure

1. Installation and commissioning completed with evidence and approval.
2. Final payment milestone becomes due; final receivable created once.
3. Customer payment link sent (or bank disbursement claimed for loan projects).
4. DISCOM inspection and net-meter installation tracked as DISCOM tasks.
5. Net meter installed: customer notified; project marked operationally complete when all handover conditions are met.
6. Payment received and reconciled: financial closure completed.

If the project is complete but payment is unpaid, status shows "Project completed — payment outstanding", never financially closed.

Flow E — Employee leave and project protection

1. Leave applied; balance checked.
2. Affected projects, tasks and customer commitments identified.
3. Impact calculated; backup suggested; handover arranged.
4. Leave approved; availability and assignments updated.
5. Affected teams notified; project health recalculated.

Flow F — Costing to profitability

1. Site-type defaults or survey measurements.
2. Equipment BOM, cable, structure, labor and travel calculated.
3. Cost, selling price and subsidy display.
4. Quotation accepted; survey verified; budget locked.
5. Procurement and material issue (including to contractors).
6. Actual material, labor, contractor and travel recorded.

7. Estimated vs actual and project profitability calculated.

Flow G — Feedback to recognition

1. Milestone completed; feedback requested and received.
2. Validity checked.
3. Employee or contractor performance updated.
4. Reward rules evaluated; recommendation generated.
5. Approval if required; reward recorded.

Flow H — Contractor work to payment

1. Contractor selected for the project; tasks auto-assigned by skill.
2. Work order generated and shared.
3. Material issued and recorded.
4. Crew completes tasks with evidence.
5. Approval creates payable at the contractor's rate.
6. Deductions, retention and TDS applied.
7. Payment approved, paid and reconciled; contractor performance updated.

Flow I — DISCOM process

1. DISCOM selected at project creation; its task set added.
2. Application and documents submitted; tasks wait on DISCOM.
3. Coordinator marks each step complete when the DISCOM finishes it.
4. Waiting time attributed to DISCOM, not the team.
5. Customer sees each step as a simple milestone.

9. UI, motion and experience

The goal is how the product feels in use: calm, fluid and premium. No generated videos.

9.1 Visual quality

- Premium, minimal and distinctive design.
- Calm colors and excellent typography.
- Consistent spacing and components.

- Strong visual hierarchy and clear status indicators.
- Beautiful responsive dashboards.
- Contextual cards and progressive disclosure.
- No cluttered screens or unnecessary tables.
- Role-specific interfaces.
- Quotation PDF designed to the same premium standard, with price, subsidy and effective cost easy to read.

9.2 Motion and interaction

- Fluid navigation, card transitions and contextual expansion.
- Micro-interactions and feedback on every important action.
- Animated data for business health, project progress, finance and capacity.
- Lightweight illustrations for success, loading, onboarding and rewards (Lottie; Rive where interactive motion adds value).
- Native mobile animations and haptics.
- 3D (Three.js/WebGL) only where it improves understanding, such as a system layout preview.

9.3 Performance rules

- No animation delays an important action.
- Heavy 3D is lazy-loaded, with lighter alternatives for low-end devices.
- Respect reduced-motion preferences.
- Optimized image delivery.
- Skeletons and progressive rendering.
- Routine navigation stays instant.

10. Platform and technical requirements

Area	Requirement
Web authentication	User ID/email and password
Mobile authentication	OTP (employees, contractors, resellers, customers)

Area	Requirement
Online/offline	Online-only; automatic retry for evidence uploads; no offline workflows or sync
Data	One shared source of truth
Permissions	Configurable roles; customer, reseller and contractor isolation
Audit	Important actions, document replacements and financial events recorded
Search	Universal search
Documents	Centralized, categorized and searchable
Notifications	Central notification and WhatsApp services
Automation	Shared workflow and rules engine
Finance	Internal double-entry ledger with payment, bank disbursement and contractor payable reconciliation
Costing and subsidy	One centralized calculation engine with configurable rule tables
Mobile	Responsive, camera-friendly and simple
Reliability	No silent transaction or data loss
Privacy	Consent capture and retention rules for location data
Security	Secure access and data isolation
Reporting	Drill-down from every metric to records

Architecture guardrails

1. Separate business logic from UI.
2. Reuse the same domain rules across web and mobile.
3. Never duplicate quotation, subsidy or costing calculations.
4. Use a shared task, workflow and notification engine.

5. Keep integrations (Razorpay, WhatsApp, banks) behind service boundaries.
6. Use proper database constraints and transactions.
7. Use idempotency for payment and workflow events.
8. Maintain audit history for sensitive operations.
9. Avoid unnecessary microservices and infrastructure complexity.
10. Prefer deterministic rules; add AI only where it creates measurable value.

11. Intentional exclusions

Excluded or limited	Decision
Offline mobile mode	Not required; only automatic upload retry
Complex document versioning	Not required; replacements kept in audit trail only
Generated videos (Remotion)	Not required
Per-kW-only cost calculation	Benchmark and default fallback only
Manual assignment of every task	Replaced by role-based assignment
OneOhm paying or deducting subsidy	Never; subsidy is displayed only
Customer notification for every task	Only explicitly enabled tasks
Customer visibility into internal tasks	Selected milestones only
Complex performance scoring	Transparent, configurable, verified metrics
AI for routine calculations	Deterministic business logic
Overloaded dashboards	Exceptions and meaningful KPIs
Excessive configuration screens	Simple guided settings with pre-built defaults
Final payment held by DISCOM steps	Final payment due at commissioning by default
Automatic financial closure on task completion	Completion and settlement stay separate
Unrestricted location tracking	Transparent, consented, policy-based tracking only

12. Delivery approach

The entire system — web app, field app, customer app, WhatsApp and the shared platform — is built and released together.

- The shared design system, data model, domain rules and engines (pricing, subsidy, task, workflow, notification, ledger) are defined first, because every app depends on them.
- Modules are built in parallel against these shared contracts.
- OneOhm's templates, rate cards, DISCOMs, subsidy rules and payment milestones are pre-configured before release.
- Release happens only when the acceptance criteria in section 13 pass end to end.

13. Acceptance criteria

The product is complete only when it demonstrates these real business outcomes, each verified by an end-to-end test scenario.

Area	Acceptance requirement
Quotation speed	A quotation with price, subsidy and effective cost is generated and shared in a few minutes without a survey
Subsidy	Correct subsidy for individual, housing society and commercial cases; ₹0 with a reason for non-DCR panels
Quote accuracy	Survey variance within tolerance is absorbed; beyond tolerance creates a revision for approval
Project creation	Conditional tasks and assignments (employees and contractors) without manual assignment of every task
DISCOM	Selecting a DISCOM adds its tasks; DISCOM waiting time is reported separately
Loan handling	Loan tasks and bank receivables appear only when a loan is selected; disbursements reconcile
Contractors	Work order, evidence, payable, deductions and payment flow end to end; contractors see only their own data
Task communication	A configured task sends the correct WhatsApp message when its trigger is reached

Area	Acceptance requirement
Task completion	Required evidence and approvals are enforced; uploads retry until they succeed
Final payment	Commissioning creates the final receivable once; net-meter completion closes the project operationally
Costing	Cable, structure, labor, travel and contractor costs are measurable and traceable
Documents	Find, download, select and print project documents easily
Leave	Project impact and handover are handled before ordinary leave approval
Engagement	Points and rewards reflect verified performance
Customer app	Customer tracks progress and subsidy status, pays, accesses documents and requests service
Reseller app	Reseller sees only their leads, projects, commissions and payments
Finance	Every online, offline and bank collection and every payout is recorded and reconciled
Owner dashboard	Owner identifies major risks and delay causes without inspecting every project
UX	New users complete common workflows without training, tested with real OneOhm staff

14. Product vision

OneOhm — one connected business. Every project under control.

- **For the owner:** one dashboard shows business health, money, projects, people, contractors and exceptions.
- **For employees and contractors:** one app shows exactly what to do, makes it easy to complete and recognizes good work.

- **For the customer:** one app or WhatsApp shows the project journey, subsidy, payments, documents and support.
- **For the system:** connect every activity, enforce business rules, trigger notifications, track responsibilities, reconcile transactions and prevent anything from being missed.

We are not building software that asks people to manage software. We are building software that manages the business processes and asks people to take only the actions that genuinely require them. Any new feature is evaluated against these goals, not added because it is technically possible.

Appendix A — Company details

Used on quotations, invoices, receipts, system-generated reports (3.18), work orders, change request letters and WhatsApp templates. Stored as an editable company profile in central configuration, never hardcoded.

Field	Value
Legal name	One Ohm Sustainable Green Energy Private Limited
Brand name	OneOhm
Full address	'OneOhm', Plot No. 93, Vasantdada Industrial Estate, Near Old RTO Office, Tal. Miraj, Dist. Sangli, Sangli - 416416
Short address	Plot No. 93, Vasantdada Industrial Estate, Sangli - 416416
GSTIN	27AAECO5032B1ZW
Email	info.oneohm@gmail.com
Contact numbers	+91 87882 75659 · +91 97979 79598 · +91 92251 76086

The full address is used on legal and statutory documents; the short address on quotations, letterheads and app screens.